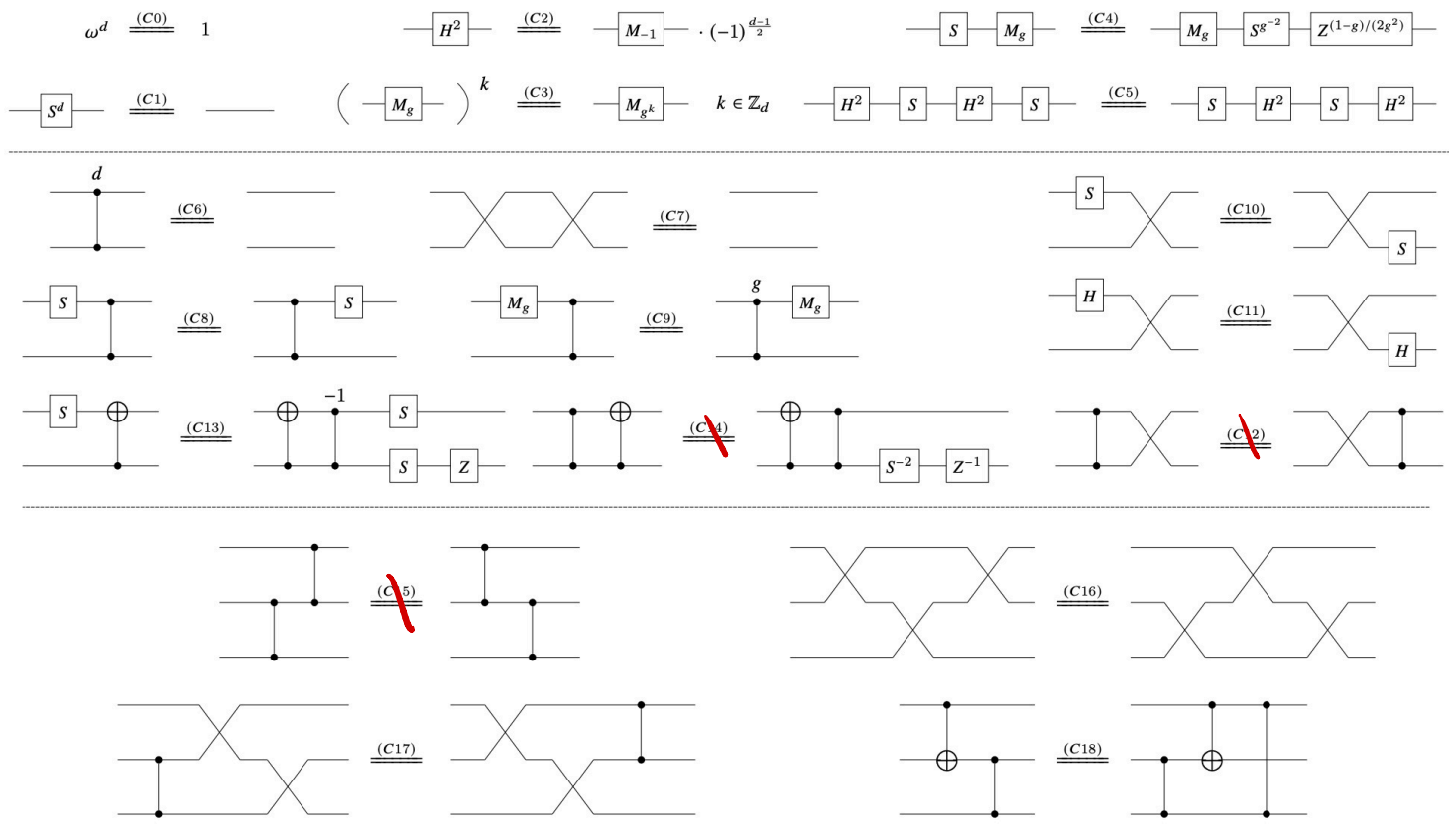


Phaseful Relation

Reduction : Part II





Lem 1 $\boxed{H} \times \text{---} \stackrel{C_{11}}{=} \text{---} \times \boxed{H}$ implies $\boxed{H} \times \text{---} \stackrel{C'_{11}}{=} \text{---} \times \boxed{H}$

Proof: $C'_{11} \cdot \text{LHS} = \boxed{H} \times \text{---} \stackrel{C_7}{=} \text{---} \times \boxed{H} \stackrel{C_{11}}{=} \text{---} \times \boxed{H} \stackrel{C_7}{=} \text{---} \times \boxed{H} = C'_{11} \cdot \text{RHS}$

D_4 : $\text{---} \times \text{---} = \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---}$

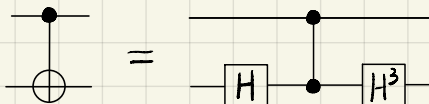
Lem 2 D_4 and C_{11} imply $\text{---} \times \text{---} \stackrel{C_{12}}{=} \text{---} \times \text{---}$

Proof: $C_{12} \cdot \text{RHS} = \text{---} \times \text{---} \stackrel{C_2}{=} \text{---} \times \begin{array}{c} \boxed{H^4} \bullet \\ \boxed{H^4} \bullet \end{array} \stackrel{C_{11}}{=} \text{---} \times \begin{array}{c} \boxed{H^3} \bullet \\ \boxed{H^3} \bullet \end{array} \stackrel{C_{11}}{=} \text{---} \times \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array}$

$\stackrel{D_4}{=} \text{---} \begin{array}{c} \boxed{H^3} \bullet \\ \boxed{H^3} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \stackrel{C_2}{=} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \begin{array}{c} \boxed{H} \bullet \\ \boxed{H} \bullet \end{array} \text{---} \stackrel{D_4}{=} \text{---} \times \text{---}$

$= C_{12} \cdot \text{LHS}$



Lem 3 D_5 : $CX :=$  soundness

Proof: $LHS |j, l\rangle = CX |j, l\rangle = |j, j+l\rangle \quad 0 \leq j, l \leq d-1$

$$\begin{aligned}
 RHS |j, l\rangle &= (I \otimes H_3) CZ (I \otimes H) |j, l\rangle \\
 &= \frac{1}{\lambda_d \sqrt{d}} (I \otimes H_3) CZ \left(|j\rangle \left(\sum_{k=0}^{d-1} \omega^{kl} |k\rangle \right) \right) = \frac{1}{\lambda_d \sqrt{d}} (I \otimes H_3) \left(\sum_{k=0}^{d-1} \omega^{kl} CZ |j, k\rangle \right) \\
 &= \frac{1}{\lambda_d \sqrt{d}} (I \otimes H_3) \left(\sum_{k=0}^{d-1} \omega^{kl} \omega^{jk} |j, k\rangle \right) = \frac{1}{\lambda_d \sqrt{d}} (I \otimes H_3) \left(\sum_{k=0}^{d-1} \omega^{k(j+l)} |j, k\rangle \right) \\
 &= \frac{1}{\lambda_d \sqrt{d}} (I \otimes H) \left(\sum_{k=0}^{d-1} \omega^{k(j+l)} |j\rangle H^2 |k\rangle \right) = \frac{1}{\lambda_d \sqrt{d}} (I \otimes H) \left(\sum_{k=0}^{d-1} \omega^{k(j+l)} |j\rangle \lambda_d^{-2} |-k\rangle \right) \\
 &= \frac{1}{\lambda_d \sqrt{d}} \lambda_d^{-2} \left(\sum_{k=0}^{d-1} \omega^{k(j+l)} |j\rangle H |-k\rangle \right) \quad \lambda_d = (-i)^{\frac{d-1}{2}} \quad \lambda_d^4 = (-i)^{2(d-1)} = (-1)^{d-1} \stackrel{d-1 \text{ is even}}{=} 1 \\
 &= \frac{1}{\lambda_d^4 d} \left(\sum_{k=0}^{d-1} \omega^{k(j+l)} |j\rangle \left(\sum_{t=0}^{d-1} \omega^{-kt} |t\rangle \right) \right) \\
 &= \frac{1}{d} \left(\sum_{t,k=0}^{d-1} \omega^{k(j+l)} \omega^{-kt} |j\rangle |t\rangle \right) = \frac{1}{d} \left(\sum_{t,k=0}^{d-1} \omega^{k(j+l-t)} |j, t\rangle \right) \\
 &= \sum_{t=0}^{d-1} \left(\frac{1}{d} \sum_{k=0}^{d-1} \omega^{k(j+l-t)} \right) |j, t\rangle = \sum_{t=0}^{d-1} C_t |j, t\rangle, \text{ where } C_t = \frac{1}{d} \sum_{k=0}^{d-1} \omega^{k(j+l-t)} \quad (1)
 \end{aligned}$$

The k -sum determines the coefficient of each basis state $|j, t\rangle$.

Let $a = j+l-t$. $C_t = \frac{1}{d} \sum_{k=0}^{d-1} \omega^{ka}$

Lemma A.1. Let ω be the primitive d -th root of unity and $k \in \mathbb{Z}_d$.

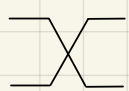
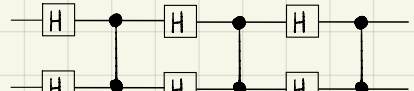
$$\sum_{\ell=0}^{d-1} \omega^{k\ell} = \begin{cases} 0, & k \not\equiv_d 0 \\ d, & k \equiv_d 0. \end{cases}$$

- When $a \equiv_d 0$ ($j+l \equiv_d t$), $C_t = 1$
- When $a \not\equiv_d 0$, $C_t = 0$

Every term $|j, t\rangle$ has coefficient 0, except the one with $t \equiv_d j+l$

$\Rightarrow (1) = RHS |j, l\rangle = |j, j+l\rangle$.

Therefore, $LHS = RHS$.

Lem 4 D4:  =  $\cdot \lambda_d^2$ soundness $H|j\rangle = \frac{1}{\lambda_d \sqrt{d}} \sum_{l=0}^{d-1} w^{jl} |l\rangle$

Proof: LHS $|j, l\rangle = \text{SWAP } |j, l\rangle = |l, j\rangle \quad 0 \leq j, l \leq d-1$.

RHS $|j, l\rangle = \lambda_d^2 \cdot CZ(H \otimes H) CZ(H \otimes H) CZ(H \otimes H) |j, l\rangle$

$$= \lambda_d^2 \cdot CZ(H \otimes H) CZ(H \otimes H) CZ \left(\frac{1}{\lambda_d^2 d} \left(\sum_{k=0}^{d-1} w^{jk} |k\rangle \right) \otimes \left(\sum_{t=0}^{d-1} w^{tl} |t\rangle \right) \right)$$

$$= \lambda_d^2 \cdot \frac{1}{\lambda_d^2 d} CZ(H \otimes H) CZ(H \otimes H) CZ \left(\sum_{k,t=0}^{d-1} w^{jk} w^{tl} |k, t\rangle \right)$$

$$= \lambda_d^2 \cdot \frac{1}{\lambda_d^2 d} CZ(H \otimes H) CZ(H \otimes H) \left(\sum_{k,t=0}^{d-1} w^{jk} w^{tl} w^{kt} |k, t\rangle \right)$$

$$= \lambda_d^2 \cdot \frac{1}{\lambda_d^4 d^2} CZ(H \otimes H) CZ \left(\sum_{k,t=0}^{d-1} w^{jk} w^{tl} w^{kt} \left(\sum_{a=0}^{d-1} w^{ak} |a\rangle \right) \otimes \left(\sum_{b=0}^{d-1} w^{bt} |b\rangle \right) \right)$$

$$= \lambda_d^2 \cdot \frac{1}{\lambda_d^4 d^2} CZ(H \otimes H) CZ \left(\sum_{a,b,k,t=0}^{d-1} w^{jk} w^{tl} w^{kt} w^{ak} w^{bt} |a, b\rangle \right)$$

$$= \lambda_d^2 \cdot \frac{1}{\lambda_d^4 d^2} CZ(H \otimes H) \left(\sum_{a,b,k,t=0}^{d-1} w^{jk} w^{tl} w^{kt} w^{ak} w^{bt} w^{ab} |a, b\rangle \right)$$

$$= \lambda_d^2 \cdot \frac{1}{\lambda_d^4 d^3} \sum_{c,d,a,b,k,t=0}^{d-1} w^{jk} w^{tl} w^{kt} w^{ak} w^{bt} w^{ab} w^{ca} w^{db} w^{cd} |c, d\rangle$$

$$= \frac{1}{d^3} \sum_{b,c,d,k,t=0}^{d-1} w^{jk+tl+kt+bt+db+cd} \sum_{a=0}^{d-1} w^{a(b+c+k)} |c, d\rangle \quad (\Delta)$$

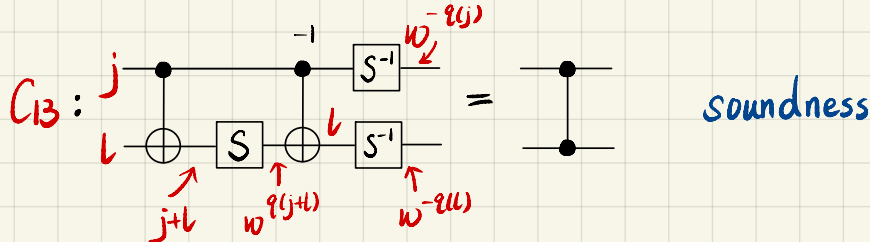
$$\sum_a w^{a(b+c+k)} = \begin{cases} 0 & b+c+k \not\equiv 0 \\ d & b+c+k \equiv 0 \Leftrightarrow b = -k-c \end{cases}$$

$$jk+tl+kt+bt+db+cd = jk+tl+kt+t(-k-c)+d(-k-c)+cd$$

$$= jk+tl+kt-kt-tc-kd-cd+cd = jk+tl-tc-kd$$

$$(\Delta) = \frac{d}{d^3} \sum_{c,d,k,t=0}^{d-1} w^{jk+tl-tc-kd} |c, d\rangle$$

Lem 5

C_{13} :  soundness

Proof: $\text{RHS } |j, l\rangle = CZ |j, l\rangle = w^{jl} |j, l\rangle \quad 0 \leq j, l \leq d-1$

$$\begin{aligned} \text{LHS } |j, l\rangle &= (S^{-1} \otimes S^{-1}) CX^{-1} (I \otimes S) CX |j, l\rangle = (S^{-1} \otimes S^{-1}) CX^{-1} (I \otimes S) |j, j+l\rangle \\ &= (S^{-1} \otimes S^{-1}) CX^{-1} w^{\frac{(j+l)(j+l-1)}{2}} |j, j+l\rangle = w^{\frac{(j+l)(j+l-1)}{2}} (S^{-1} \otimes S^{-1}) |j, l\rangle \\ &= w^{\frac{(j+l)(j+l-1)}{2}} w^{-\frac{j(j-1)}{2}} w^{-\frac{l(l-1)}{2}} |j, l\rangle = w^{jl} |j, l\rangle = \text{RHS } |j, l\rangle \end{aligned}$$

$$\frac{(j+l)(j+l-1)}{2} - \frac{j(j-1)}{2} - \frac{l(l-1)}{2} = \frac{(j+l)^2 - (j+l) - j(j-1) - l(l-1)}{2} = \frac{j^2 + l^2 + 2jl - j^2 - j - j^2 + j - l^2 + l - l^2 + l}{2} = jl$$

A bilinear CZ phase means the phase factor of the qudit CZ gate is given by a bilinear function of the two computation basis states $|j, l\rangle \mapsto w^{jl} |j, l\rangle \quad j, l \in \mathbb{Z}_d$ □

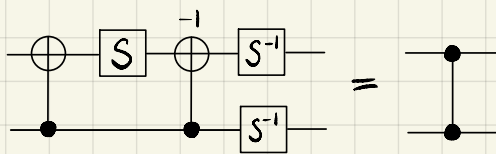
jl is bilinear because it's linear in j when l is fixed $(j_1 + j_2)l = j_1 l + j_2 l$
it's linear in l when j is fixed $j(l_1 + l_2) = j l_1 + j l_2$

$q(j)$ is the shorthand for the quadratic phase polynomial of the qudit S gate.

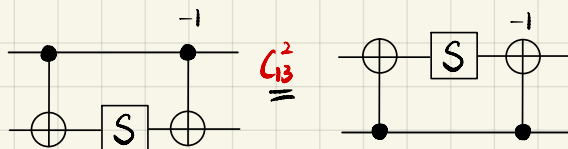
$$q(j) = \frac{j(j-1)}{2} \quad \text{with } j \in \mathbb{Z}_d. \quad S|j\rangle = w^{q(j)} |j\rangle$$

$$q(j+l) - q(j) - q(l) = jl$$

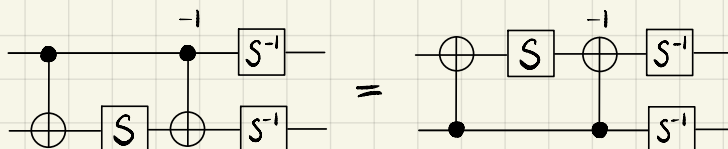
Cor 5 C_{13} implies C'_{13} :



Lem 6 C_{13} and C'_{13} imply



Proof: By C_{13} and C'_{13}



By C_1 , we complete the proof.

R_9^{33} :

R_9^{47} :

C_{13}^1 :

C_{13}^2 :

T_3 : $Z = S^{-1} K S K$, $K = H^2$

C_{13}^1 :

Lem 7 C_{13}^1 implies

Proof:

R_9^{47} R_9^{33} :

R_9^{33} :

□

R_9^{47} :

Quantum circuit diagram for R_9^{47} . The circuit consists of two horizontal lines. The top line has a box labeled CZ^a . The bottom line has a box labeled H^2 . These two boxes are connected by a vertical line, indicating a controlled operation. To the right of this is an equals sign, followed by another circuit diagram. In this second diagram, the bottom line has a box labeled H^2 , and the top line has a box labeled CZ^{-a} . They are also connected by a vertical line.

C_{13}^2 :

Diagrammatic equation for C_{13}^2 :

Left side: A box labeled S is between two circles on the bottom wire. The top wire has a dot at the start and a dot labeled -1 at the end.

Right side: A box labeled S is between two circles on the top wire. The bottom wire has a dot at the start and a dot labeled -1 at the end.

The two sides are separated by an equals sign.

C'_3 :



The diagram shows the transformation of circuit C'_3 . On the left, a circuit with two horizontal lines. The top line has a circle with a plus sign, followed by a box labeled 'S', and then a circle with a minus sign. The bottom line has a solid black dot at the position of the first plus sign and another solid black dot at the position of the second minus sign. An equals sign follows. On the right, the transformed circuit is shown. The top line has a solid black dot at the first position, followed by a box labeled 'S'. The bottom line has a solid black dot at the first position, followed by a box labeled 'S'.

Proof:

Diagrammatic proof showing the commutativity of C_{13} and C_8 . The diagram illustrates the transformation of a circuit involving S and S^{-1} gates and phase shifts, demonstrating that the operations commute.

A block diagram of a discrete-time system. The input signal enters a summing junction (circle with a plus sign). The output of this junction goes through a block labeled S^{-1} . This signal then enters a second summing junction (circle with a plus sign). The output of the second junction goes through a block labeled S . The output of the S block is fed back through a block labeled S^{-1} to the second summing junction. There is a red handwritten label (1) next to the feedback path.

Note that

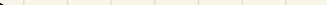
and

Then

The diagram shows a sequence of four quantum circuit diagrams connected by equals signs and a red double-headed arrow, illustrating the simplification of a multi-controlled NOT gate. The first circuit has two qubits. The top qubit has a multi-controlled NOT gate controlled by S and S^{-1} . The bottom qubit has a multi-controlled NOT gate controlled by S and K . The second circuit shows the top qubit with a NOT gate and a CNOT controlled by the bottom qubit. The third circuit shows the top qubit with a NOT gate and a CNOT controlled by the bottom qubit, with a red double-headed arrow indicating equivalence. The fourth circuit shows the top qubit with a NOT gate and a CNOT controlled by the bottom qubit, with a NOT gate on the bottom qubit.

Hence, (1) =

The diagram illustrates the reduction of a feedback system. The left side shows the original system with two parallel paths. The top path consists of a summing junction, a block labeled -1 , a block labeled s^{-1} , and another summing junction, followed by a block labeled S . The bottom path consists of a summing junction, a block labeled S^{-1} , and another summing junction. The two paths are connected at three points. An orange box highlights the first two summing junctions and the -1 block. The right side shows the reduced system. The top path consists of a summing junction, a block labeled s^{-1} , and a summing junction, followed by a block labeled S . The bottom path consists of a summing junction, a block labeled K , a block labeled s^{-1} , a block labeled K , and a summing junction, followed by a block labeled S^{-1} . The two paths are connected at two points. An orange box highlights the first two summing junctions and the K and s^{-1} blocks in the bottom path.

$$=$$

, since $KS^{-1}KS^{-1} = KS^{-1}KS^{-2} = Z^{-1}S^{-2} = S^{-2}Z^{-1}$.

$$C_{15}: \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array}$$

$$C_{16}: \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array}$$

$$C_{17}: \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array}$$

$$C_{18}: \begin{array}{c} \oplus \\ \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \oplus \\ \bullet \end{array}$$

$$R_9^{15}: \begin{array}{c} \boxed{CZ^a} \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{CZ^{ag}} \\ \bullet \\ \bullet \end{array}$$

$$R_9^{14}: \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{CZ^a} \\ \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{CZ^{ag^{-1}}} \\ \bullet \\ \bullet \end{array}$$

Lem 9 C_{18} implies C_{15}

Proof: Up to qubit wire permutation, we derive the following equation.

$$C_{15}: \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} = \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array}$$

$$\begin{aligned} C_{15} \text{ RHS} &= \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \stackrel{C_6}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \stackrel{C_{18}}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \\ &= \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \stackrel{R_9^{15}}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \stackrel{C_{18}}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \\ &\stackrel{R_9^{14}}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \stackrel{C_6}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \stackrel{C_{18}}{=} \begin{array}{c} -1 \\ \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \\ &= C_{15} \text{ LHS} \end{aligned}$$

□